

```
$sudo adduser -ingroup hadoop hduser #this will add hduser
```

Networking: Before we continue to build our Hadoop cluster, we need to make sure that all nodes can communicate with each other. According to the block diagram (shown in Figure 1) we will have one master node (*hdnamenode*) and three slave nodes (*hddatanode1*, *hddatanode2* and *hddatanode3*) as shown below:

```
$sudo vi /etc/hosts
172.15.10.2 hdnamenode
172.15.10.3 hddatanode1
172.15.10.4 hddatanode2
172.15.10.5 hddatanode3
```



Note: If there are more datanodes (slaves), these need to be updated in */etc/hosts*.

Enabling SSH: Hadoop requires SSH access to manage all its nodes. *hduser* on *hdnamenode* needs to connect to itself and to the datanodes (in our case, *hddatanode1*, *hddatanode2* and *hddatanode3*) via passwordless SSH login. Installing Hadoop in a passwordless manner is not recommended, but in this case it is needed to unlock the system without user interaction to avoid entering the passphrase every time Hadoop interacts with its nodes.

```
hduser@hdnamenode:~$ su -i hduser
hduser@hdnamenode:~$ ssh-keygen -t rsa -P ""
hduser@hdnamenode:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
hduser@hdnamenode:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub hddatanode1
hduser@hdnamenode:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub hddatanode2
hduser@hdnamenode:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub hddatanode3
```

Installation

Download the latest stable Hadoop version, which is 2.6.0, from <http://hadoop.apache.org/releases.html> and extract the contents from the Hadoop package to the location of your choice.



Note: Make sure to change the owner of all the files to the *hduser* and *hadoop* groups.

```
$cd /usr/local
$sudo tar xzf Hadoop-2.6.0.tar.gz
$sudo mv Hadoop.2.6.0 Hadoop
$sudo chown -R hduser:hadoop hadoop
```

Add the lines below at the end of *~/.bashrc*

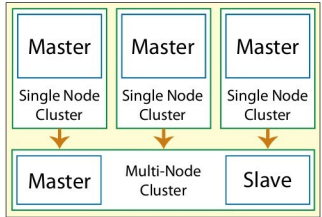


Figure 1: Architecture of a Hadoop multi-node cluster

```
#HADOOP VARIABLES START
export JAVA_HOME="/usr/lib/jvm/jdk1.7.0_67/"
export PATH="$PATH:$JAVA_HOME/bin"
export HADOOP_INSTALL="/usr/local/hadoop/etc/hadoop"
export PATH="$PATH:$HADOOP_INSTALL/bin"
export PATH="$PATH:$HADOOP_INSTALL/sbin"
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export YARN_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_INSTALL/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_INSTALL/lib"
$source ~/.bashrc # refresh the bashrc file
```

Edit */usr/local/hadoop/etc/hadoop/hadoop-env.sh*

```
# set to the root of your Java installation
export JAVA_HOME="/usr/lib/jvm/java-7-openjdk-i386"
# assuming installation directory is /usr/local/hadoop
export HADOOP_PREFIX="/usr/local/hadoop"
```

Configuration details

Let's configure the following files in order to build a Hadoop multi-mode cluster -- *masters*, *slaves*, *core-site.xml*, *mapred-site.xml* and *hdfs-site.xml*.

Masters: In Hadoopmaster node, navigate to */usr/local/hadoop/etc/hadoop* folder and edit the *masters* file as shown below:

```
hduser@hdnamenode:~$vi /usr/local/hadoop/etc/hadoop/masters
hdnamenode
```

Slaves: Add all the slave nodes, one per line, as shown below:

```
hduser@hdnamenode:~$vi /usr/local/hadoop/etc/hadoop/slaves
hdnamenode
hddatanode1
```